English translation prepared with ChatGPT (OpenAI, GPT-6), 26 October 2026, for translation of the Bulgarian text into English and LaTeX formatting. The Bulgarian originals remain the reference; translation may contain errors.

Probability and Statistics, Software Engineering, Year 3, 2020/2021 Written exam, 27.01.2021

Your grade is 2 plus the number of points earned. Time allowed: 3 hours. Good luck.
Throughout, we work on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

Problem 1. (1 point) An urn contains balls numbered from 1 to 10 inclusive. Three balls are drawn without replacement. Define the following events concerning the balls drawn:

1. $A =$*the largest number is 6*;
2. $B =$*at least two balls have even numbers*;
3. $C =$*the sum of the numbers is divisible by 3.*

Find the probabilities of these events and determine whether any are independent. What is $\mathbb{P}(A|B)$? What is the expected number of even-numbered balls drawn if sampling is with replacement?

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Let $c = \binom{10}{3} = 120$. Then

$$\mathbb{P}(A) = \frac{\binom{5}{2}}{c} = \frac{1}{12}, \quad \mathbb{P}(B) = \frac{\binom{5}{3} + 5\binom{5}{2}}{c} = \frac{1}{2}\,.$$

For $\mathbb{P}(C)$, either enumerate the combinations directly, or observe that the possible combinations modulo 3 are three equal residues or $0+1+2$. Hence

$$\mathbb{P}(C) = \frac{3 \cdot 4 \cdot 3 + 2 \cdot \binom{3}{3} + \binom{4}{3}}{c} = \frac{7}{20}\,.$$

Direct calculation/enumeration gives

$$\mathbb{P}(A \cap B) = \frac{\binom{2}{2} + \binom{3}{1} \cdot 2}{c} = \frac{7}{120}, \mathbb{P}(A \cap C) = \frac{3}{120}, \mathbb{P}(B \cap C) = \frac{21}{120}\,.$$

No pair is independent, and

$$\mathbb{P}(A|B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)} = \frac{7}{60}\,.$$

With replacement, the number of even-numbered balls drawn has distribution $Bin(3, 1/2)$, so the required expectation is $3/2$.

Problem 2. (1 point) Two instruments measure the concentration of dust particles in the air. Suppose the true concentration is x g/m^3. The reading of the first instrument is normally distributed with mean x and standard deviation (σ) $0.05x$. The second reading is also normally distributed, with mean x and standard deviation $0.1x$. Which instrument would you use? For each instrument, what is the probability of an error greater than $0.1x$?

Someone decides to use the arithmetic mean of the two readings. If the measurements are independent, what is the probability of an error greater than $0.1x$ with this procedure?

• The instrument with smaller variance is preferable. Denote the readings of the first and second instruments by $\xi_1 \sim N(x, (0.05x)^2)$ and $\xi_2 \sim N(x, (0.1x)^2)$. The required probabilities are

$$\begin{aligned}\mathbb{P}(|\xi_1 - x| > 0.1x) &= \mathbb{P}(\xi_1 < 0.9x) + \mathbb{P}(\xi_1 > 1.1x) = 2\mathbb{P}(\xi_1 < 0.9x) \\ &= 2\mathbb{P}\left(N(0,1) < \frac{0.9x - x}{0.1x}\right) = 2\mathbb{P}(N(0,1) < -1) \\ &\approx 0.3174\,.\end{aligned}$$

and similarly

$$\mathbb{P}(|\xi_2 - x| > 0.1x) = 2\mathbb{P}(N(0,1) < -2) \approx 0.0456\,.$$

If we define $\xi_3 = 1/2(\xi_1 + \xi_2)$, then $\xi_3 \sim N(x, 1/4(0.05^2 + 0.1^2)x^2)$ and

$$\mathbb{P}(|\xi_3 - x| > 0.1x) = 2\mathbb{P}\left(N(0,1) < \frac{-0.1x}{\sqrt{1/4(0.05^2 + 0.1^2)x^2}}\right) \approx 2\mathbb{P}(N(0,1) < -1.79) \approx 0.0734\,.$$

Problem 3. (1 point) Let ξ and η be independent random variables, $\xi \sim Exp(2)$ and $\eta \sim U(0,3)$, i.e.

$$f_\xi(x) = \begin{cases} 2e^{-2x} & \text{if } x > 0 \\ 0 & \text{otherwise} \end{cases}; \quad f_\eta(x) = \begin{cases} \frac{1}{3} & \text{if } 0 < x < 3 \\ 0 & \text{otherwise.} \end{cases}$$

Find the correlation of ξ and η, $P(\xi < \eta)$ and the density of ξ/η.

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The correlation is 0 because ξ and η are independent. Considering $0 < \eta < 3$ and $0 < \xi < \eta$, we obtain

$$\mathbb{P}(\xi < \eta) = \int_0^3 \int_0^y 2e^{-2x} \cdot \frac{1}{3} \,\mathrm{d}x \,\mathrm{d}y = \frac{1}{6}\left(5 + \frac{1}{e^6}\right) \approx 0.8334\,.$$

Note that independence gives $f_{\xi,\eta}(x,y) = 2/3 \cdot e^{-2x}\mathbb{1}_{x>0,0<y<3}$. Define $X = \xi/\eta$ and $Y = \eta$. The inverse transformation is $\xi = XY$ and $\eta = Y$, with Jacobian Y. Therefore

$$f_{X,Y}(x,y) = y f_{\xi,\eta}(xy,y) = y\frac{2}{3}e^{-2xy}\mathbb{1}_{xy>0,0<y<3}\,.$$

Thus, for $x > 0$,

$$\begin{aligned} f_X(x) &= \int_\infty^\infty f_{X,Y}(x,y)\,\mathrm{d}y = \frac{2}{3}\int_0^3 ye^{-2xy}\,\mathrm{d}y = \frac{1}{6x^2}\int_0^3 -2xy \cdot e^{-2xy}\,\mathrm{d}(-2xy) \\ &= \frac{1}{6x^2}\int_0^{-6x} t \cdot e^t\,\mathrm{d}t = \frac{e^{-6x}(-6x-1)+1}{6x^2}\,, \end{aligned}$$

as required. The last result also gives another way to calculate the probability requested earlier:

$$\mathbb{P}(\xi < \eta) = \mathbb{P}\left(\frac{\xi}{\eta} < 1\right) = \int_0^1 f_X(x)\,\mathrm{d}x = \int_0^1 \frac{e^{-6x}(-6x-1)+1}{6x^2}\,\mathrm{d}x$$

Problem 4. (1 point) Let ξ and η have joint density

$$f_{\xi,\eta}(x,y) = \begin{cases} cx^3y^2 & \text{if } 0 < x,y < 2 \\ 0 & \text{otherwise.} \end{cases}$$

Find the constant c, the correlation of ξ and η, and $\mathbb{E}(\xi|\eta = 1)$.

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$$1 = c\int_0^2\int_0^2 x^3y^2\,\mathrm{d}x\,\mathrm{d}y = c\frac{32}{3}\,.$$

Hence $c = 3/32 = 0.09735$. For $x \in (0,2)$,

$$f_\xi(x) = \int_0^2 cx^3y^2\,\mathrm{d}y = cx^3\frac{8}{3} = \frac{1}{4}x^3\,.$$

Therefore

$$\mathbb{E}\xi = \frac{1}{4}\int_0^2 x^4\,\mathrm{d}x = \frac{8}{5} = 1.6, \quad D\xi = \frac{1}{4}\int_0^2 x^5\,\mathrm{d}x - \frac{64}{25} = \frac{8}{75} \approx 0.1067\,.$$

Similarly, for $y \in (0,2)$,

$$f_\eta(y) = c\int_0^2 x^3y^2\,\mathrm{d}x = \frac{3}{8}y^2\,,$$

$$\mathbb{E}\eta = \frac{3}{8}\int_0^2 y^3\,\mathrm{d}y = \frac{3}{2} = 1.5, \quad D\eta = \frac{3}{8}\int_0^2 y^4\,\mathrm{d}y - \frac{9}{4} = \frac{3}{20} = 0.15\,.$$

Next,

$$\mathbb{E}\xi\eta = \frac{3}{32}\int_0^2\int_0^2 xy(x^3y^2)\,\mathrm{d}x\,\mathrm{d}y = \frac{12}{5} = 2.4$$

Therefore

$$Cor(\xi,\eta) = \frac{\mathbb{E}\xi\eta - \mathbb{E}\xi\mathbb{E}\eta}{\sqrt{D\xi D\eta}} = 0\,.$$

For the last question,

$$\mathbb{E}(\xi|\eta = 0.5) = \int_0^2 x f_{\xi|\eta}(x|y=1)\,\mathrm{d}x = \int_0^2 x\frac{f_{\xi,\eta}(x,1)}{f_\eta(1)}\,\mathrm{d}x = \int_0^2 x\frac{x^3}{4}\,\mathrm{d}x = \frac{8}{5} = 1.6\,.$$